

Auditory Notifications in Everyday Contexts: A User Study on Smart Glasses

Start of Block: Intro

Intro 0 **Welcome to our study on Smart Glasses and Audio Notifications!** We are investigating how future smart glasses can intelligently adapt audio notifications based on your current environment. **Important Information:** 🕶️ **No Prior Experience Needed:** You do **not** need to own, or have ever used, smart glasses to participate! We will explain everything you need to know. ⌚ **Duration:** The study takes approximately **15 minutes**. 🎧 **Requirements:** You must use a **Desktop or Laptop computer** (no mobile phones) and **wear Headphones or Earbuds**. **Good English proficiency is required.** 🔒 **Privacy & Anonymity:** Your participation is entirely voluntary. All data is collected completely anonymously and will be used strictly for academic purposes. It is not possible to trace the data back to you. You can stop the survey at any time without giving a reason by closing your browser window. 💰 **Compensation:** Either **0.25 Study Participation Hours (MMI Points) per commenced 15-minute interval** or **monetary compensation** at a rate of **€12/hour, prorated** based on your participation time. For any questions, please contact:
simon.roedig@campus.lmu.de
Supervisor: albrecht.schmidt@ifi.lmu.de, LMU Munich

End of Block: Intro

Start of Block: Informed Consent



Informed Consent **Informed Consent of Participation** You are invited to participate in this online study, conducted by Simon Roedig as part of a Master's Thesis in Human-Computer Interaction. The research is supervised by Laura Schütz and Thomas Weber, under the direction of Prof. Dr. Albrecht Schmidt (Media Informatics Group, LMU Munich). Please note: Your participation is voluntary. The online study will last approximately 15 minutes. We will record personal demographics (Age, gender, country of origin, native language, English proficiency, hearing assessment, musical background). No names or IP addresses will be linked to the survey data. We will record interaction data (e.g., clicks, timings) during the session. All data you provide will be published completely anonymized; it will not include your name and cannot be associated with your identity. You may choose to receive compensation for your

participation in the form of either:

Study Participation Hours

(MMI Points) (0.25 Points per commenced 15-minute interval), or

Monetary compensation at a rate of €12 per hour, prorated based on participation time.

Compensation details will be provided at the end of the study. Any contact information collected for compensation purposes will be stored separately from your study responses and will not be linked to your survey data.

1. Purpose and Goal of this Research The purpose of this research is to investigate how future smart glasses can intelligently adapt auditory notifications based on a user's physical, social, and acoustic environment. The goal is to understand user preferences for audio notification modalities (sounds vs. speech) and delivery timing across various simulated everyday situations. The anonymized results of this research may be presented at scientific meetings or published in scientific journals.

2. Participation and Withdrawal Your participation in this online study is strictly voluntary. You will be one of approximately 30–50 people surveyed for this research. You may withdraw and discontinue participation at any time simply by closing your browser window. You may also refuse to answer any questions you do not want to answer. At any time and without giving any reason, you can notify us that you want to withdraw your consent (GDPR Art. 7(3)). If you withdraw before completing the survey, any partial data collected will be deleted.

3. Procedure As part of this study, you will:

- Answer basic demographic and background questions.
- Familiarize yourself with three different audio notification types using your headphones.
- Immerse yourself in six visual/auditory scenarios and evaluate the notifications for each.
- Answer concluding questions regarding your overall preferences and system design.

4. Risks and Benefits There are no anticipated risks associated with this online study. Discomforts or inconveniences will be minor and are unlikely to happen. You will not directly benefit from participation, but your input will help advance knowledge in the field of Context-Aware Human-Computer Interaction.

5. Data Protection and Confidentiality The legal basis for processing personal data is the participant's consent pursuant to Art. 6(1)(a) GDPR. The GDPR guarantees a set of rights to the data subjects, including:

- The right to access your personal data (GDPR Art. 15).
- The right to correct inaccurate personal data (GDPR Art. 16).
- The right to have your personal data deleted (GDPR Art. 17).
- The right to limit or object to the processing of your personal data (GDPR Art. 18 & 21).

We will record personal demographics and interaction logs during the online study. All data you provide will be stored and published strictly anonymized. We will remove any personal information that could identify you before publishing the data, ensuring no one can trace the information back to you. Your anonymized data will be stored for up to 10 years (in accordance with good scientific practice guidelines by the DFG). Anonymized data collected may be shared publicly in scientific repositories (e.g., Open Science Framework) for open science purposes.

6. Identification of Investigators If you have any questions or concerns about the research or your rights as a participant, please feel free to contact:

- Conductor: Simon Roedig (simon.roedig@campus.lmu.de)
- Supervisors: Laura Schütz (laura.schuetz@tum.de) & Thomas Weber (thomas.weber@ifi.lmu.de)
- Department Head: Prof. Dr. Albrecht Schmidt (LMU Munich)

7. Informed Consent and Agreement This consent form will be retained securely and in compliance with the GDPR for no longer than necessary.

Participation in this study requires agreement to all consent statements below. If you do not wish to provide consent, please close the survey and do not proceed:

☐

I understand the explanation provided to me. I was able to save a copy of this form. I reached out to the listed researchers and have had all my questions answered to my satisfaction. Therefore, I voluntarily agree to participate in this online study. (1)

☐

I voluntarily consent to my data being recorded and subsequently processed in line with the GDPR. I have been informed about the consequences of withdrawing my consent. (2)

End of Block: Informed Consent

Start of Block: Intro 2

Intro 1 **Smart glasses** (such as the Meta Ray-Ban) are wearable devices equipped with 📷 cameras, 🎤 microphones, and 🔊 **discreet private speakers built into the frames** (similar to earbuds). They enable users to conveniently capture photos and videos, interact with an integrated AI assistant through voice commands, and **receive audio notifications through the integrated speakers without disturbing those nearby.**

Intro 2 **In this Online-Study:** You will **evaluate three different methods** by which smart glasses can **deliver notifications of new WhatsApp messages through audio**. You will answer questions, compare approaches, and rate your experience across various scenarios.

Desktop Laptop To participate in this study, you must be **using a desktop or laptop computer** (not a mobile phone). Are you using a desktop or laptop computer?

☐ Yes, I am using a desktop or laptop computer. (1)

☐ No, I am using a mobile device. (2)

Headphones This study requires the **use of headphones/earbuds**. Are you wearing headphones/earbuds? Please put them on now.

- ☐ Yes, I am wearing headphones. (1)
- ☐ No, I am not wearing headphones. (2)
-

Audio Check  **Listen to this sound to adjust your volume.** Adjust your device volume until this sound is at a comfortable but clearly audible level with your headphones/earbuds. **Do NOT change the volume again during the study.** Dein Browser untersttzt das Audio-

Element nicht. (Note: Adjust the volume setting of your computer/laptop, NOT the volume of this audio clip) Qualtrics.SurveyEngine.addOnload(function () { var audio = document.getElementById("stimulusAudio"); if (!audio) return; // Force full volume initially audio.volume = 1; audio.muted = false; // Prevent participant from changing volume audio.addEventListener("volumechange", function () { setTimeout(function () { audio.volume = 1; audio.muted = false; }, 0); }); // Re-apply whenever playback starts audio.addEventListener("play", function () { audio.volume = 1; audio.muted = false; }); });

Headphone Check  Please listen to the following audio clip and indicate whether the sound is played from the **left channel**, **right channel**, or **both channels**.

Qualtrics.SurveyEngine.addOnload(function () { var audio = document.getElementById("stimulusAudio2"); if (!audio) return; // Force full volume initially audio.volume = 1; audio.muted = false; // Prevent participant from changing volume audio.addEventListener("volumechange", function () { setTimeout(function () { audio.volume = 1; audio.muted = false; }, 0); }); // Re-apply whenever playback starts audio.addEventListener("play", function () { audio.volume = 1; audio.muted = false; }); });

- ☐ Left Channel (1)
- ☐ Right Channel (2)
- ☐ Both Channels (3)
- ☐ (Unsure) (4)

End of Block: Intro 2

Start of Block: Demographics and Background



Age What is your age?

Skip To: End of Survey If Condition: What is your age? Is Less Than 18. Skip To: End of Survey.

Gender What is your gender?

- ☐ Male (1)
 - ☐ Female (2)
 - ☐ Non-binary / third gender (3)
 - ☐ Prefer not to say (4)
-

Culture What is your country of origin?



Language What is your native/primary language?

English Skill Please rate your English proficiency for understanding spoken English

	Cannot understand at all (1)	Understand very little (2)	Understand somewhat (3)	Understand moderately well (4)	Understand well (5)	Understand very well (6)	Understand perfectly (7)
English proficiency (1)	☐	☐	☐	☐	☐	☐	☐

Impairment Have you ever been diagnosed with a hearing impairment?

- ☐ No (1)
- ☐ Yes (Describe) (2) _____
-

Musical Background How would you describe your **musical background** and experience?

- ☐ I rarely/never listen to music and do not play an instrument. (1)
- ☐ I regularly listen to music, but do not play an instrument. (2)
- ☐ I know how to play an instrument (or sing), but rarely do. (3)
- ☐ I regularly play an instrument (or sing) as a hobby / perform publicly. (4)
- ☐ I am a professionally trained musician / singer. (5)
-

My Notif Settings On your primary **smartphone**, **what** is your standard **notification setting** for incoming private **messages** (e.g., **WhatsApp**)?

- ☐ No notification at all (Muted / Do Not Disturb) (1)
- ☐ Visual only (Screen lights up or shows banner, no sound/vibration) (2)
- ☐ Visual and Vibration (Screen lights up + phone vibrates) (3)
- ☐ Visual and Audio (Screen lights up + sound plays) (4)
- ☐ Visual, Vibration, and Audio (Screen lights up + phone vibrates + sound plays) (5)

My Notif Settings Why What is the primary **reason** for your **chosen notification setting**?
(Select multiple)

- ☐ Social Etiquette: To avoid disturbing or annoying people around me. (1)
- ☐ Personal Focus: To avoid distracting myself or breaking my concentration. (2)
- ☐ Privacy: So people around me don't know when I receive a message. (3)
- ☐ Awareness: To ensure I don't miss any messages (e.g., needing sound/vibration to notice them). (4)
- ☐ Habit: It's just my default setting / I never bothered to change it. (5)
- ☐ Other: (6) _____

Glass Knowledge How familiar are you with smart (AI) glasses (e.g. Meta Ray-Ban)?

	Not familiar at all (1)	Slightly familiar (2)	Moderately familiar (3)	Familiar (4)	Very familiar (5)
Familiarity (1)	☐	☐	☐	☐	☐

Glass Experience Have you ever used smart (AI) glasses (e.g. Meta Ray-Ban)?

	Never (1)	Tried once or twice (2)	Used occasionally (3)	Used regularly (4)	Use frequently (5)
Experience (1)	☐	☐	☐	☐	☐

Which Notif Auditory Imagine you are wearing **smart glasses with private speakers** (no integrated display) in your everyday life. Which of the following (smartphone) notifications would you **actually** want to **receive AUDITORILY** 🗣️ through the glasses? (Only you can hear them)

- ☐ Phone Calls (1)
 - ☐ Direct Instant Messages (e.g., WhatsApp, iMessage) (2)
 - ☐ Group Messages (3)
 - ☐ Emails (4)
 - ☐ Calendar / Reminder Alerts (5)
 - ☐ Social Media interactions (e.g., likes, new followers) (6)
 - ☐ Navigation / Directions (7)
 - ☐ None of the above (8)
 - ☐ Others: (9) _____
-

Notif ImprtantUrgent Please indicate how strongly you agree or disagree with the following statements.

	Strongly disagree (1)	Disagree (2)	Somewhat disagree (3)	Neither agree nor disagree (4)	Somewhat agree (5)	Agree (6)	Strongly agree (7)
I would only want my smart glasses to notify me auditorily about messages (e.g. from WhatsApp) that are highly important. (1)	☐	☐	☐	☐	☐	☐	☐
I would only want my smart glasses to notify me auditorily about messages (e.g. from WhatsApp) that are highly urgent. (2)	☐	☐	☐	☐	☐	☐	☐

End of Block: Demographics and Background

Start of Block: Familiarization

Intro Versions Imagine you have received a new WhatsApp message from **LILLY (please imagine Lilly as a very close friend of yours)**. She wrote: **"When are you here?"** This new message is both **URGENT** and **IMPORTANT** for you! Please listen to the following three versions your smart glasses can use to inform you of this new WhatsApp message:

🔔 Version 1: 🧐 Version 2: 🧐🧐 Version 3: ⚠️ They are **ONLY** audible to you via the smart glasses' speakers. **No one around you can hear them.** ⚠️ The smart

glasses we consider do **not have an integrated display**. You only hear them auditorily. (To possibly access more information about the WhatsApp message, you could either check your phone or interact with the smart glasses using voice commands or a dedicated button.) → During this study, you will then hear these three versions multiple times in different scenarios. → We always consider the same WhatsApp message from Lilly that is URGENT and IMPORTANT to you. → Note that the voice in the two spoken notifications is only an example and could also vary in tone, style, or speaker. Focus on the content rather than the voice characteristics.

```
Qualtrics.SurveyEngine.addOnload(function () {
  var audios = document.querySelectorAll(".stimAudio");
  if (!audios.length) return;
  audios.forEach(function(audio) {
    // Force full volume
    audio.volume = 1;
    audio.muted = false;
    // Prevent volume changes
    audio.addEventListener("volumechange", function () {
      setTimeout(function () {
        audio.volume = 1;
        audio.muted = false;
      }, 0);
    });
    // Only one audio may play at a time
    audio.addEventListener("play", function () {
      audios.forEach(function(otherAudio) {
        if (otherAudio !== audio) {
          otherAudio.pause();
          otherAudio.currentTime = 0;
        }
      });
    });
    // Re-enforce volume
    audio.volume = 1;
    audio.muted = false;
  });
});
```

Sender Check Check: Which **two** versions informed you about the sender's name?

☐

 Version 1 (1)

☐

 Version 2 (2)

☐

 Version 3 (3)

WhatsApp In this study we are focusing on **WhatsApp notifications**. 🛎 Version 1's sound is the typical WhatsApp notification sound. **How is your prior experience with this notification sound?**

	Strongly disagree (1)	Disagree (2)	Somewhat disagree (3)	Neither agree nor disagree (4)	Somewhat agree (5)	Agree (6)	Strongly agree (7)
I am familiar with this sound. (1)	☐	☐	☐	☐	☐	☐	☐
I generally associate this sound with WhatsApp. (2)	☐	☐	☐	☐	☐	☐	☐
I specifically associate this sound with the arrival of a new WhatsApp message. (3)	☐	☐	☐	☐	☐	☐	☐

Preference Imagine you are wearing smart glasses with private speakers (only you can hear the audio). You receive **this urgent and important notification from Lilly**. Generally, **independent of your current situation and task, how would you prefer to be notified?**

- ☐ 🛎 Version 1 (1)
- ☐ 🗣 Version 2 (2)
- ☐ 🗣 🗣 Version 3 (3)
- ☐ I don't want to be notified auditorily about this notification (4)

Preference Text **Briefly explain** why you prefer your **chosen version**. (Answer in English or German)

Scenes Inappropriate Can you think of a specific real-world scenario where this **preferred notification version would become inappropriate** and you would **want the smart glasses to use one of the other versions instead?** (Answer in English or German)

Speech Opinion Please indicate to what extent you **agree with the following statements.**

	Strongly disagree (1)	Disagree (2)	Somewhat disagree (3)	Neither agree nor disagree (4)	Somewhat agree (5)	Agree (6)	Strongly agree (7)
If my smart glasses suddenly read a message out loud to me instead of just playing a standard "ding" sound, my first assumption would be that the message is highly urgent or important. (1)	☐	☐	☐	☐	☐	☐	☐
If I were wearing smart glasses with private speakers in my everyday life, I would find it acceptable for the system to read notifications out loud without my explicit request. (2)	☐	☐	☐	☐	☐	☐	☐

Speech Opinion 2 Which of the following **factors influenced** your answer to the **previous question**? (Select all that apply)

- ☐ Privacy (others cannot hear it anymore) (1)
 - ☐ Speech is easier to understand (2)
 - ☐ Reduces need to look at a device (3)
 - ☐ Reduces need to talk (back) with the smart glasses (4)
 - ☐ Too intrusive (5)
 - ☐ I prefer control over when notifications speak (6)
 - ☐ I don't trust automatic speech output (7)
 - ☐ Depends on context (8)
 - ☐ Depends on the notification/message (9)
 - ☐ Other (please specify): (10)
-

Maximum Length What is the **maximum length of a spoken notification** you would find **acceptable** in most situations? (Can Select Multiple)

- ☐ Very short (1 sentence / headline only) (1)
- ☐ Short (2–3 sentences) (2)
- ☐ Medium (up to ~1 minute spoken content) (3)
- ☐ Long (no strong limit) (4)
- ☐ It depends on the situation (5)
- ☐ It depends on the message/notification (6)
- ☐ Other: (7) _____

Summary Acceptance Please indicate to what extent you agree with the following statement.

	Strongly disagree (1)	Disagree (2)	Somewhat disagree (3)	Neither agree nor disagree (4)	Somewhat agree (5)	Agree (6)	Strongly agree (7)
I find it highly acceptable if the system automatically summarized LONG WhatsApp messages instead of reading the exact text. (1)	☐	☐	☐	☐	☐	☐	☐

End of Block: Familiarization

Start of Block: Instructor

Master Instruction You will now experience **6 different everyday situations**. For every scenario, please assume the following: 🕶️ You are **wearing smart glasses**. 🎧 The audio **notifications are delivered privately to your ears. No one else around you can hear them.**

💬 You just received the **URGENT and IMPORTANT WhatsApp message** ("*When are you here?*") from **Lilly (a very close friend of yours)**. Your **Task for Each Scenario**: 1. **Read the short description** of the situation. 2. Carefully **view the image and listen to all 3 notification** versions. 3. Feel free to replay the audio versions as often as needed **WHILE looking at the image**, until you **feel adequately immersed in the scenario**. 4. **Answer the questions** that follow.

End of Block: Instructor

Start of Block: Scene A (Home Alone Computer)

Instruction A Imagine you are in the following situation: ➡️ You are **working alone at your home desk in a quiet and familiar environment. You are currently engaged in a cognitively demanding task: Writing code and solving a complex problem. You are wearing smart glasses. Notifications are delivered privately through speakers and are not audible to others.** ⬅️ 💬 Lilly wrote you an **urgent and important message**: "*When are you here?*"

Task A 🔔 Version 1: 🗣️ Version 2: 🗣️🗣️ Version 3:
Qualtrics.SurveyEngine.addOnLoad(function () { var audios =
document.querySelectorAll(".qmedia"); if (!audios) return; audios.forEach(function(audio)
{ // 🔊 enforce default volume audio.volume = 1; // 🔒 lock volume (prevent
user changes) audio.addEventListener("volumechange", function () {
setTimeout(function () { audio.volume = 1; }, 0); }); // 🔊 only one
audio plays at a time audio.addEventListener("play", function () {
audios.forEach(function(otherAudio) { if (otherAudio !== audio) {
otherAudio.pause(); otherAudio.currentTime = 0; } });
audio.volume = 1; }); });

A. Appropriateness Overall, how **appropriate** is this notification in the given situation?

	Completely inappropriate (1)	Inappropriate (2)	Somewhat inappropriate (3)	Neither appropriate nor inappropriate (4)	Somewhat appropriate (5)	Appropriate (6)	Completely appropriate (7)
 Version 1 (1)	☐	☐	☐	☐	☐	☐	☐
 Version 2 (2)	☐	☐	☐	☐	☐	☐	☐
 Version 3 (3)	☐	☐	☐	☐	☐	☐	☐

A. Detectability How easily were you able to **detect** the notification in this environment?

	Very difficult to detect (1)	Difficult to detect (2)	Somewhat difficult to detect (3)	Neither easy nor difficult to detect (4)	Somewhat easy to detect (5)	Easy to detect (6)	Very easy to detect (7)
 Version 1 (1)	☐	☐	☐	☐	☐	☐	☐
 Version 2 (2)	☐	☐	☐	☐	☐	☐	☐
 Version 3 (3)	☐	☐	☐	☐	☐	☐	☐

A. Disruption How **disruptive** was this notification to your ongoing activity in this scenario?

	Not disruptive at all (1)	Slightly disruptive (2)	Somewhat disruptive (3)	Moderately disruptive (4)	Disruptive (5)	Very disruptive (6)	Extremely disruptive (7)
 Version 1 (1)	☐	☐	☐	☐	☐	☐	☐
 Version 2 (2)	☐	☐	☐	☐	☐	☐	☐
 Version 3 (3)	☐	☐	☐	☐	☐	☐	☐

A. Social How **socially acceptable** did it **feel** to receive this type of notification in this situation?

	Completely unacceptable (1)	Unacceptable (2)	Somewhat unacceptable (3)	Neither acceptable nor unacceptable (4)	Somewhat acceptable (5)	Acceptable (6)	Completely Acceptable (7)
 Version 1 (1)	☐	☐	☐	☐	☐	☐	☐
 Version 2 (2)	☐	☐	☐	☐	☐	☐	☐
 Version 3 (3)	☐	☐	☐	☐	☐	☐	☐

A. Overall Overall, which audio notification would you **prefer** in this situation?

☐ 🔔 Version 1 (1)

☐ 🗣️ Version 2 (2)

☐ 🗣️ 🗣️ Version 3 (3)

☐ No audio notification at all (4)

A. Why Text Briefly share the **reason** behind your **preference**: (Answer in English or German)

A. Overall Why Which of the following **factors** most strongly **influenced your choice** of audio notification in this scenario? (Select multiple)

☐ Acoustic Environment (e.g., the level of background noise, music, or other people talking) (1)

☐ Focus & Mental Effort (e.g., needing to concentrate on my current task without being interrupted) (2)

☐ Social Appropriateness (e.g., the presence of others, social etiquette, or not wanting to appear distracted) (3)

☐ Physical Safety & Awareness (e.g., needing to pay visual or mental attention to my physical surroundings, like traffic) (4)

☐ Desire for Information (e.g., wanting to know the exact sender or content immediately without having to use my hands/phone) (5)

☐ Other (please specify): (6)

A. Timing **When** would you want an audio notification to play?

☐ Immediately (1)

☐ When I briefly pause while writing / thinking (2)

☐ After I finish the current task (e.g., finish writing (part of the) code) (3)

☐ Later, when I am done working (4)

A. Timing Follow If you chose to **delay** the notification in the previous question, **which version** would you prefer the system to play **when that later time arrives**?

- ☐ 🔔 Version 1 (1)
- ☐ 🗣️ Version 2 (2)
- ☐ 🗣️ 🗣️ Version 3 (3)
- ☐ Not applicable (I chose "Immediately" in the previous question). (4)
-

A. Why Timing Text Briefly share the **reason** behind your **preferred timing** (and, **if you chose to delay, why** you prefer **that specific notification version** when the time arrives)

End of Block: Scene A (Home Alone Computer)

Start of Block: Scene B (Team Meeting)

Instruction B **Instruction:** Imagine you are in the following situation: ➡️ **You are in an office meeting room with colleagues. The meeting is lively, people are speaking, asking questions, and you are actively listening and contributing. There is continuous background speech from the discussion. You are wearing smart glasses. Notifications are delivered privately through speakers and are not audible to others.** ⬅️ 🗣️ Lilly wrote you an **urgent and important message**: *"When are you here?"*

Task B 🔔 Version 1: 🗣️ Version 2: 🗣️ 🗣️ Version 3:

```
Qualtrics.SurveyEngine.addOnLoad(function () {    var audios =  
document.querySelectorAll(".qmedia");    if (!audios) return;    audios.forEach(function(audio)  
{        // 🗣️ enforce default volume        audio.volume = 1;        // 🔒 lock volume (prevent  
user changes)        audio.addEventListener("volumechange", function () {
```

```

setTimeout(function () {          audio.volume = 1;          }, 0);    });    // 🔊 only one
audio plays at a time          audio.addEventListener("play", function () {
audios.forEach(function(otherAudio) {          if (otherAudio !== audio) {
otherAudio.pause();          otherAudio.currentTime = 0;          }          });
audio.volume = 1;    });    });

```

B. Appropriateness Overall, how **appropriate** is this notification in the given situation?

	Completely inappropriate (1)	Inappropriate (2)	Somewhat inappropriate (3)	Neither appropriate nor inappropriate (4)	Somewhat appropriate (5)	Appropriate (6)	Completely appropriate (7)
 Version 1 (1)	☐	☐	☐	☐	☐	☐	☐
 Version 2 (2)	☐	☐	☐	☐	☐	☐	☐
 Version 3 (3)	☐	☐	☐	☐	☐	☐	☐

B. Detectability How easily were you able to **detect** the notification in this environment?

	Very difficult to detect (1)	Difficult to detect (2)	Somewhat difficult to detect (3)	Neither easy nor difficult to detect (4)	Somewhat easy to detect (5)	Easy to detect (6)	Very easy to detect (7)
 Version 1 (1)	☐	☐	☐	☐	☐	☐	☐
 Version 2 (2)	☐	☐	☐	☐	☐	☐	☐
 Version 3 (3)	☐	☐	☐	☐	☐	☐	☐

B. Disruption How **disruptive** was this notification to your ongoing activity in this scenario?

	Not disruptive at all (1)	Slightly disruptive (2)	Somewhat disruptive (3)	Moderately disruptive (4)	Disruptive (5)	Very disruptive (6)	Extremely disruptive (7)
 Version 1 (1)	☐	☐	☐	☐	☐	☐	☐
 Version 2 (2)	☐	☐	☐	☐	☐	☐	☐
 Version 3 (3)	☐	☐	☐	☐	☐	☐	☐

B. Social How **socially acceptable** did it **feel** to receive this type of notification in this situation?

	Completely unacceptable (1)	Unacceptable (2)	Somewhat unacceptable (3)	Neither acceptable nor unacceptable (4)	Somewhat acceptable (5)	Acceptable (6)	Completely Acceptable (7)
 Version 1 (1)	☐	☐	☐	☐	☐	☐	☐
 Version 2 (2)	☐	☐	☐	☐	☐	☐	☐
 Version 3 (3)	☐	☐	☐	☐	☐	☐	☐

B. Overall Overall, which audio notification would you **prefer** in this situation?

☐ 🔔 Version 1 (1)

☐ 🗣️ Version 2 (2)

☐ 🗣️ 🗣️ Version 3 (3)

☐ No audio notification at all (4)

B. Why Text Briefly share the **reason** behind your **preference**: (Answer in English or German)

B. Overall Why Which of the following **factors** most strongly **influenced your choice** of audio notification in this scenario? (Select multiple)

☐ Acoustic Environment (e.g., the level of background noise, music, or other people talking) (1)

☐ Focus & Mental Effort (e.g., needing to concentrate on my current task without being interrupted) (2)

☐ Social Appropriateness (e.g., the presence of others, social etiquette, or not wanting to appear distracted) (3)

☐ Physical Safety & Awareness (e.g., needing to pay visual or mental attention to my physical surroundings, like traffic) (4)

☐ Desire for Information (e.g., wanting to know the exact sender or content immediately without having to use my hands/phone) (5)

☐ Other (please specify): (6)

B. Timing **When** would you want an audio notification to play?

☐ Immediately (1)

☐ When I am not speaking and there is a natural break in conversation (2)

☐ After the meeting segment I am currently involved in ends (3)

☐ Later, when the meeting is over (4)

B. Timing Follow If you chose to **delay** the notification in the previous question, **which version** would you prefer the system to play **when that later time arrives**?

- ☐ 🔔 Version 1 (1)
- ☐ 🗣️ Version 2 (2)
- ☐ 🗣️ 🗣️ Version 3 (3)
- ☐ Not applicable (I chose "Immediately" in the previous question). (4)
-

B. Why Timing Text Briefly share the **reason** behind your **preferred timing** (and, **if you chose to delay, why** you prefer **that specific notification version** when the time arrives)

End of Block: Scene B (Team Meeting)

Start of Block: Scene C (Home Music)

Instruction C **Instruction:** Imagine you are in the following situation: ➡️ **You are at home, alone. Loud music is playing through a stereo music system in the room. You are relaxing. Your phone is laying on the couch table. You are wearing smart glasses. Notifications are delivered privately through speakers and are not audible to others.** ⬅️
💬 **Lilly** wrote you an **urgent and important message**: "*When are you here?*"

Task C 🔔 Version 1: 🗣️ Version 2: 🗣️ 🗣️ Version 3:
Qualtrics.SurveyEngine.addOnload(function () { var audios =
document.querySelectorAll(".qmedia"); if (!audios) return; audios.forEach(function(audio)
{ // 🗣️ enforce default volume audio.volume = 1; // 🔒 lock volume (prevent
user changes) audio.addEventListener("volumechange", function () {

```

setTimeout(function () {          audio.volume = 1;          }, 0);    });    // 🔊 only one
audio plays at a time          audio.addEventListener("play", function () {
audios.forEach(function(otherAudio) {          if (otherAudio !== audio) {
otherAudio.pause();          otherAudio.currentTime = 0;          }          });
audio.volume = 1;    });    });

```

C. Appropriateness Overall, how **appropriate** is this notification in the given situation?

	Completely inappropriate (1)	Inappropriate (2)	Somewhat inappropriate (3)	Neither appropriate nor inappropriate (4)	Somewhat appropriate (5)	Appropriate (6)	Completely appropriate (7)
 Version 1 (1)	☐	☐	☐	☐	☐	☐	☐
 Version 2 (2)	☐	☐	☐	☐	☐	☐	☐
 Version 3 (3)	☐	☐	☐	☐	☐	☐	☐

C. Detectability How easily were you able to **detect** the notification in this environment?

	Very difficult to detect (1)	Difficult to detect (2)	Somewhat difficult to detect (3)	Neither easy nor difficult to detect (4)	Somewhat easy to detect (5)	Easy to detect (6)	Very easy to detect (7)
 Version 1 (1)	☐	☐	☐	☐	☐	☐	☐
 Version 2 (2)	☐	☐	☐	☐	☐	☐	☐
 Version 3 (3)	☐	☐	☐	☐	☐	☐	☐

C. Disruption How **disruptive** was this notification to your ongoing activity in this scenario?

	Not disruptive at all (1)	Slightly disruptive (2)	Somewhat disruptive (3)	Moderately disruptive (4)	Disruptive (5)	Very disruptive (6)	Extremely disruptive (7)
 Version 1 (1)	☐	☐	☐	☐	☐	☐	☐
 Version 2 (2)	☐	☐	☐	☐	☐	☐	☐
 Version 3 (3)	☐	☐	☐	☐	☐	☐	☐

C. Social How **socially acceptable** did it **feel** to receive this type of notification in this situation?

	Completely unacceptable (1)	Unacceptable (2)	Somewhat unacceptable (3)	Neither acceptable nor unacceptable (4)	Somewhat acceptable (5)	Acceptable (6)	Completely Acceptable (7)
 Version 1 (1)	☐	☐	☐	☐	☐	☐	☐
 Version 2 (2)	☐	☐	☐	☐	☐	☐	☐
 Version 3 (3)	☐	☐	☐	☐	☐	☐	☐

C. Overall Overall, which audio notification would you **prefer** in this situation?

☐ 🔔 Version 1 (1)

☐ 🗣️ Version 2 (2)

☐ 🗣️ 🗣️ Version 3 (3)

☐ No audio notification at all (4)

C. Why Text Briefly share the **reason** behind your **preference**: (Answer in English or German)

C. Overall Why Which of the following **factors** most strongly **influenced your choice** of audio notification in this scenario? (Select multiple)

☐ Acoustic Environment (e.g., the level of background noise, music, or other people talking) (1)

☐ Focus & Mental Effort (e.g., needing to concentrate on my current task without being interrupted) (2)

☐ Social Appropriateness (e.g., the presence of others, social etiquette, or not wanting to appear distracted) (3)

☐ Physical Safety & Awareness (e.g., needing to pay visual or mental attention to my physical surroundings, like traffic) (4)

☐ Desire for Information (e.g., wanting to know the exact sender or content immediately without having to use my hands/phone) (5)

☐ Other (please specify): (6)

C. Timing **When** would you want an audio notification to play?

☐ Immediately (1)

☐ When I am between songs / during a natural pause in music (2)

☐ When I switch from relaxing to another task (e.g. tidying room). Music continues. (3)

☐ Later, when I am done with music and relaxing (4)

C. Timing Follow If you chose to **delay** the notification in the previous question, **which version** would you prefer the system to play **when that later time arrives**?

- ☐ 🔔 Version 1 (1)
- ☐ 🗣️ Version 2 (2)
- ☐ 🗣️ 🗣️ Version 3 (3)
- ☐ Not applicable (I chose "Immediately" in the previous question). (4)

C. Why Timing Text Briefly share the **reason** behind your **preferred timing** (and, **if you chose to delay, why** you prefer **that specific notification version** when the time arrives)

End of Block: Scene C (Home Music)

Start of Block: Attention Check

D. Attention Check To ensure our survey is working correctly, please select '**Somewhat disagree**' for this question.

	Strongly disagree (1)	Disagree (2)	Somewhat disagree (3)	Neither agree nor disagree (4)	Somewhat agree (5)	Agree (6)	Strongly agree (7)
Select (1)	☐	☐	☐	☐	☐	☐	☐

End of Block: Attention Check

Start of Block: Scene D (Coffee Bar)

Instruction D **Instruction:** Imagine you are in the following situation: ➡ **You are sitting across your good friend REBECCA at a café. You are in the middle of telling her a story. There is light background noise (other quiet conversations, coffee machine), but you are actively engaged in a conversation. You are wearing smart glasses. Notifications are delivered privately through speakers and are not audible to others.** ⬅️ 🗨️ **Lilly wrote you an urgent and important message: "When are you here?"**

Task D 🛎 Version 1: 🗣 Version 2: 🗣🗣 Version 3:

```
Qualtrics.SurveyEngine.addOnload(function () {
  var audios = document.querySelectorAll(".qmedia");
  if (!audios) return;
  audios.forEach(function(audio) {
    // 🗣 enforce default volume
    audio.volume = 1;
    // 🔒 lock volume (prevent user changes)
    audio.addEventListener("volumechange", function () {
      setTimeout(function () {
        audio.volume = 1;
      }, 0);
    });
    // 🗣 only one audio plays at a time
    audio.addEventListener("play", function () {
      audios.forEach(function(otherAudio) {
        if (otherAudio !== audio) {
          otherAudio.pause();
          otherAudio.currentTime = 0;
        }
      });
    });
    audio.volume = 1;
  });
});
```

D. Appropriateness Overall, how **appropriate** is this notification in the given situation?

	Completely inappropriate (1)	Inappropriate (2)	Somewhat inappropriate (3)	Neither appropriate nor inappropriate (4)	Somewhat appropriate (5)	Appropriate (6)	Completely appropriate (7)
🛎 Version 1 (1)	☐	☐	☐	☐	☐	☐	☐
🗣 Version 2 (2)	☐	☐	☐	☐	☐	☐	☐
🗣🗣 Version 3 (3)	☐	☐	☐	☐	☐	☐	☐

D. Detectability How easily were you able to **detect** the notification in this environment?

	Very difficult to detect (1)	Difficult to detect (2)	Somewhat difficult to detect (3)	Neither easy nor difficult to detect (4)	Somewhat easy to detect (5)	Easy to detect (6)	Very easy to detect (7)
 Version 1 (1)	☐	☐	☐	☐	☐	☐	☐
 Version 2 (2)	☐	☐	☐	☐	☐	☐	☐
 Version 3 (3)	☐	☐	☐	☐	☐	☐	☐

D. Disruption How **disruptive** was this notification to your ongoing activity in this scenario?

	Not disruptive at all (1)	Slightly disruptive (2)	Somewhat disruptive (3)	Moderately disruptive (4)	Disruptive (5)	Very disruptive (6)	Extremely disruptive (7)
 Version 1 (1)	☐	☐	☐	☐	☐	☐	☐
 Version 2 (2)	☐	☐	☐	☐	☐	☐	☐
 Version 3 (3)	☐	☐	☐	☐	☐	☐	☐

D. Social How **socially acceptable** did it **feel** to receive this type of notification in this situation?

	Completely unacceptable (1)	Unacceptable (2)	Somewhat unacceptable (3)	Neither acceptable nor unacceptable (4)	Somewhat acceptable (5)	Acceptable (6)	Completely Acceptable (7)
 Version 1 (1)	☐	☐	☐	☐	☐	☐	☐
 Version 2 (2)	☐	☐	☐	☐	☐	☐	☐
 Version 3 (3)	☐	☐	☐	☐	☐	☐	☐

D. Overall Overall, which audio notification would you **prefer** in this situation?

- ☐  Version 1 (1)
- ☐  Version 2 (2)
- ☐  Version 3 (3)
- ☐ No audio notification at all (4)

D. Why Text Briefly share the **reason** behind your **preference**: (Answer in English or German)

D. Overall Why Which of the following **factors** most strongly **influenced your choice** of audio notification in this scenario? (Select multiple)

☐ Acoustic Environment (e.g., the level of background noise, music, or other people talking) (1)

☐ Focus & Mental Effort (e.g., needing to concentrate on my current task without being interrupted) (2)

☐ Social Appropriateness (e.g., the presence of others, social etiquette, or not wanting to appear distracted) (3)

☐ Physical Safety & Awareness (e.g., needing to pay visual or mental attention to my physical surroundings, like traffic) (4)

☐ Desire for Information (e.g., wanting to know the exact sender or content immediately without having to use my hands/phone) (5)

☐ Other (please specify): (6)

D. Timing **When** would you want an audio notification to play?

☐ Immediately (1)

☐ When I pause while speaking to listen to my friend (2)

☐ After I finish telling my current story / thought (3)

☐ Later, after the conversation ends (4)

D. Timing Follow If you chose to **delay** the notification in the previous question, **which version** would you prefer the system to play **when that later time arrives**?

- ☐ 🔔 Version 1 (1)
- ☐ 🗣️ Version 2 (2)
- ☐ 🗣️ 🗣️ Version 3 (3)
- ☐ Not applicable (I chose "Immediately" in the previous question). (4)
-

D. Why Timing Text Briefly share the **reason** behind your **preferred timing** (and, **if you chose to delay, why** you prefer **that specific notification version** when the time arrives)

End of Block: Scene D (Coffee Bar)

Start of Block: Scene E (Cycle City)

Instruction E **Instruction:** Imagine you are in the following situation: ➡️ **You are cycling alone through a busy city street. There is traffic noise, pedestrians, and urban sounds. You are focused on safely navigating the environment. You are wearing smart glasses.**

Notifications are delivered privately through speakers and are not audible to others. ⬅️

💬 **Lilly** wrote you an **urgent and important message**: *"When are you here?"*

Task E 🔔 Version 1: 🗣️ Version 2: 🗣️ 🗣️ Version 3:

```
Qualtrics.SurveyEngine.addOnload(function () {    var audios =  
document.querySelectorAll(".qmedia");    if (!audios) return;    audios.forEach(function(audio)  
{        // 🗣️ enforce default volume        audio.volume = 1;        // 🔒 lock volume (prevent  
user changes)        audio.addEventListener("volumechange", function () {
```

```

setTimeout(function () {          audio.volume = 1;          }, 0);    });    // 🔊 only one
audio plays at a time          audio.addEventListener("play", function () {
audios.forEach(function(otherAudio) {          if (otherAudio !== audio) {
otherAudio.pause();          otherAudio.currentTime = 0;          }          });
audio.volume = 1;    });    });

```

E. Appropriateness Overall, how **appropriate** is this notification in the given situation?

	Completely inappropriate (1)	Inappropriate (2)	Somewhat inappropriate (3)	Neither appropriate nor inappropriate (4)	Somewhat appropriate (5)	Appropriate (6)	Completely appropriate (7)
 Version 1 (1)	☐	☐	☐	☐	☐	☐	☐
 Version 2 (2)	☐	☐	☐	☐	☐	☐	☐
 Version 3 (3)	☐	☐	☐	☐	☐	☐	☐

E. Detectability How easily were you able to **detect** the notification in this environment?

	Very difficult to detect (1)	Difficult to detect (2)	Somewhat difficult to detect (3)	Neither easy nor difficult to detect (4)	Somewhat easy to detect (5)	Easy to detect (6)	Very easy to detect (7)
 Version 1 (1)	☐	☐	☐	☐	☐	☐	☐
 Version 2 (2)	☐	☐	☐	☐	☐	☐	☐
 Version 3 (3)	☐	☐	☐	☐	☐	☐	☐

E. Disruption How **disruptive** was this notification to your ongoing activity in this scenario?

	Not disruptive at all (1)	Slightly disruptive (2)	Somewhat disruptive (3)	Moderately disruptive (4)	Disruptive (5)	Very disruptive (6)	Extremely disruptive (7)
 Version 1 (1)	☐	☐	☐	☐	☐	☐	☐
 Version 2 (2)	☐	☐	☐	☐	☐	☐	☐
 Version 3 (3)	☐	☐	☐	☐	☐	☐	☐

E. Social How **socially acceptable** did it **feel** to receive this type of notification in this situation?

	Completely unacceptable (1)	Unacceptable (2)	Somewhat unacceptable (3)	Neither acceptable nor unacceptable (4)	Somewhat acceptable (5)	Acceptable (6)	Completely Acceptable (7)
 Version 1 (1)	☐	☐	☐	☐	☐	☐	☐
 Version 2 (2)	☐	☐	☐	☐	☐	☐	☐
 Version 3 (3)	☐	☐	☐	☐	☐	☐	☐

E. Overall Overall, which audio notification would you **prefer** in this situation?

☐ 🔔 Version 1 (1)

☐ 🗣️ Version 2 (2)

☐ 🗣️ 🗣️ Version 3 (3)

☐ No audio notification at all (4)

E. Why Text Briefly share the **reason** behind your **preference**: (Answer in English or German)

E. Overall Why Which of the following **factors** most strongly **influenced your choice** of audio notification in this scenario? (Select multiple)

☐ Acoustic Environment (e.g., the level of background noise, music, or other people talking) (1)

☐ Focus & Mental Effort (e.g., needing to concentrate on my current task without being interrupted) (2)

☐ Social Appropriateness (e.g., the presence of others, social etiquette, or not wanting to appear distracted) (3)

☐ Physical Safety & Awareness (e.g., needing to pay visual or mental attention to my physical surroundings, like traffic) (4)

☐ Desire for Information (e.g., wanting to know the exact sender or content immediately without having to use my hands/phone) (5)

☐ Other (please specify): (6)

E. Timing **When** would you want an audio notification to play?

☐ Immediately (1)

☐ When I am not actively crossing a street or making a decision about movement (2)

☐ After I reach a safe stopping point (e.g., at a traffic light) (3)

☐ After I have finished cycling / reached my destination (4)

E. Timing Follow If you chose to **delay** the notification in the previous question, **which version** would you prefer the system to play **when that later time arrives**?

- ☐ 🔔 Version 1 (1)
- ☐ 🗣️ Version 2 (2)
- ☐ 🗣️ 🗣️ Version 3 (3)
- ☐ Not applicable (I chose "Immediately" in the previous question). (4)
-

E. Why Timing Text Briefly share the **reason** behind your **preferred timing** (and, **if you chose to delay, why** you prefer **that specific notification version** when the time arrives)

End of Block: Scene E (Cycle City)

Start of Block: Scene F (Grocery Shopping)

F. Instruction **Instruction:** Imagine you are in the following situation: ➡️ **You are grocery shopping in a supermarket, carrying several bags. Your hands are occupied. There is moderate ambient store noise. Other shoppers are around but you are not interacting with anyone specific. You are wearing smart glasses. Notifications are delivered privately through speakers and are not audible to others.** ⬅️ 🗣️ Lilly wrote you an **urgent and important message**: *"When are you here?"*

F. Task 🔔 Version 1: 🗣️ Version 2: 🗣️ 🗣️ Version 3:

```
Qualtrics.SurveyEngine.addOnLoad(function () {    var audios =  
document.querySelectorAll(".qmedia");    if (!audios) return;    audios.forEach(function(audio)  
{        // 🗣️ enforce default volume        audio.volume = 1;        // 🔒 lock volume (prevent  
user changes)        audio.addEventListener("volumechange", function () {
```

```

setTimeout(function () {          audio.volume = 1;          }, 0);    });    // 🔊 only one
audio plays at a time          audio.addEventListener("play", function () {
audios.forEach(function(otherAudio) {          if (otherAudio !== audio) {
otherAudio.pause();          otherAudio.currentTime = 0;          }          });
audio.volume = 1;    });    });

```

F. Appropriateness Overall, how **appropriate** is this notification in the given situation?

	Completely inappropriate (1)	Inappropriate (2)	Somewhat inappropriate (3)	Neither appropriate nor inappropriate (4)	Somewhat appropriate (5)	Appropriate (6)	Completely appropriate (7)
 Version 1 (1)	☐	☐	☐	☐	☐	☐	☐
 Version 2 (2)	☐	☐	☐	☐	☐	☐	☐
 Version 3 (3)	☐	☐	☐	☐	☐	☐	☐

F. Detectability How easily were you able to **detect** the notification in this environment?

	Very difficult to detect (1)	Difficult to detect (2)	Somewhat difficult to detect (3)	Neither easy nor difficult to detect (4)	Somewhat easy to detect (5)	Easy to detect (6)	Very easy to detect (7)
 Version 1 (1)	☐	☐	☐	☐	☐	☐	☐
 Version 2 (2)	☐	☐	☐	☐	☐	☐	☐
 Version 3 (3)	☐	☐	☐	☐	☐	☐	☐

F. Disruption How **disruptive** was this notification to your ongoing activity in this scenario?

	Not disruptive at all (1)	Slightly disruptive (2)	Somewhat disruptive (3)	Moderately disruptive (4)	Disruptive (5)	Very disruptive (6)	Extremely disruptive (7)
 Version 1 (1)	☐	☐	☐	☐	☐	☐	☐
 Version 2 (2)	☐	☐	☐	☐	☐	☐	☐
 Version 3 (3)	☐	☐	☐	☐	☐	☐	☐

F. Social How **socially acceptable** did it **feel** to receive this type of notification in this situation?

	Completely unacceptable (1)	Unacceptable (2)	Somewhat unacceptable (3)	Neither acceptable nor unacceptable (4)	Somewhat acceptable (5)	Acceptable (6)	Completely Acceptable (7)
 Version 1 (1)	☐	☐	☐	☐	☐	☐	☐
 Version 2 (2)	☐	☐	☐	☐	☐	☐	☐
 Version 3 (3)	☐	☐	☐	☐	☐	☐	☐

F. Overall Overall, which audio notification would you **prefer** in this situation?

☐ 🔔 Version 1 (1)

☐ 🗣️ Version 2 (2)

☐ 🗣️ 🗣️ Version 3 (3)

☐ No audio notification at all (4)

F. Why Text Briefly share the **reason** behind your **preference**: (Answer in English or German)

F. Overall Why Which of the following **factors** most strongly **influenced your choice** of audio notification in this scenario? (Select multiple)

☐ Acoustic Environment (e.g., the level of background noise, music, or other people talking) (1)

☐ Focus & Mental Effort (e.g., needing to concentrate on my current task without being interrupted) (2)

☐ Social Appropriateness (e.g., the presence of others, social etiquette, or not wanting to appear distracted) (3)

☐ Physical Safety & Awareness (e.g., needing to pay visual or mental attention to my physical surroundings, like traffic) (4)

☐ Desire for Information (e.g., wanting to know the exact sender or content immediately without having to use my hands/phone) (5)

☐ Other (please specify): (6)

F. Timing **When** would you want an audio notification to play?

☐ Immediately (1)

☐ Between picking up items on my shopping list (2)

☐ Once I have collected all items and start walking to the checkout (3)

☐ After I am done shopping (4)

F. Timing Follow If you chose to **delay** the notification in the previous question, **which version** would you prefer the system to play **when that later time arrives**?

- ☐ 🛎 Version 1 (1)
- ☐ 👤 Version 2 (2)
- ☐ 👤 👤 Version 3 (3)
- ☐ Not applicable (I chose "Immediately" in the previous question). (4)
-

F. Why Timing Text Briefly share the **reason** behind your **preferred timing** (and, **if you chose to delay, why** you prefer **that specific notification version** when the time arrives)

End of Block: Scene F (Grocery Shopping)

Start of Block: Last Questions

Almost Done **Almost done. Just a few last questions to go!**

Realistic Feeling Looking back at the 6 scenarios, **how easy was it for you to mentally put yourself in those situations and imagine that you were actually there?**

	Very difficult (1)	Difficult (2)	Somewhat difficult (3)	Neither easy nor difficult (or just "Neutral") (4)	Somewhat easy (5)	Easy (6)	Very easy (7)
Ease of Immersion (1)	☐	☐	☐	☐	☐	☐	☐

Preference Matrix In the **previous** scenarios, the **WhatsApp message was an urgent & important question** requiring your attention ("When are you here?"). If you were in a **typical, everyday environment**, which **audio notification would you prefer** for the following types of messages?

	Mute (Do not notify me auditorily at all)  (1)	Sound Only 'Version 1'  (2)	Speech 'Version 2' (Sender)  (3)	Speech 'Version 3' (Sender + Full/Summarized Content)   (4)
A simple acknowledgment (e.g., "Ok", "Sounds good", "👍") (1)	☐	☐	☐	☐
A routine everyday update (e.g., "I finally finished reading that book you recommended.") (2)	☐	☐	☐	☐
Great news that doesn't require immediate action (e.g., "I just got the job!" or "We passed the exam!") (3)	☐	☐	☐	☐
Time-sensitive situations requiring immediate action (e.g., "I'm waiting outside, come down!" or "Are you on your way?") (4)	☐	☐	☐	☐

Delay Matrix Imagine your smart glasses **detect that you are in a highly focused or busy situation** (e.g., deep in work, conversation, or navigating traffic). ➡ The system decides to **delay** the notification to avoid interrupting you. Based on the **type of message**, **what is the MAXIMUM delay** you would find acceptable before the system must notify you?

	No delay: Must interrupt me immediately, regardless of my situation. (1)	Brief delay (Seconds): Can wait for a quick safe moment (e.g., until I finish crossing the street, silence). (2)	Moderate delay (Minutes): Can wait until I finish my current task (e.g., finishing an email or a conversation). (3)	Indefinite delay (Hours): Can be withheld entirely until I manually check my phone or ask the device. (4)
A simple acknowledgment (e.g., "Ok", "Sounds good", "👍") (1)	☐	☐	☐	☐
A routine everyday update (e.g., "I finally finished reading that book you recommended.") (2)	☐	☐	☐	☐
Great news that doesn't require immediate action (e.g., "I just got the job!" or "We passed the exam!") (3)	☐	☐	☐	☐
Time-sensitive situations requiring immediate action (e.g., "I'm waiting outside, come down!" or "Are you on your way?") (4)	☐	☐	☐	☐

System Opinion Imagine an **AI-based** system integrated into your AR glasses that: - **Monitors** incoming WhatsApp messages - Decides **whether** and **when** to notify you based on your

current situation - Adapts the **notification style** (sound only 🛎️, short speech 🗣️, or full speech 🗣️🗣️) depending on your context

	Strongly disagree (1)	Disagree (2)	Somewhat disagree (3)	Neither agree nor disagree (4)	Somewhat agree (5)	Agree (6)	Strongly agree (7)
Such a system would be useful to me. (1)	☐	☐	☐	☐	☐	☐	☐
I would trust such a system to make the right choice most of the time. (2)	☐	☐	☐	☐	☐	☐	☐
I would be concerned about the system choosing the wrong type at the wrong moment. (3)	☐	☐	☐	☐	☐	☐	☐

System Opinion Why Do you have any **concerns** about such a system? (**Optional**, answer in English or German)

Final We have reached the **end of the study**. If you have any **additional comments, thoughts, or impressions** you can share them here: (**Optional**, answer in English or German)

End of Block: Last Questions
